# NINAD NAIK

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## EDUCATION

**Bachelor of Engineering in Information Technology** 2021 - 2025
University of Mumbai, Thadomal Shahani Engineering College (TSEC), Mumbai, India CGPA: **8.54 / 10**

## WORK EXPERIENCE

**Associate Backend Developer**, *Commotion* (acq. by Tata Communications) (gocommotion.com) Jan 2025 - Sep 2026

- Promoted from Backend Developer Intern (January 2025 - June 2025)
- Engineered **low-latency backend architectures** using NestJS and FastAPI, utilizing targeted database indexes and query optimization to handle **high-concurrency enterprise workloads**.
- Developed **MCP and A2A orchestration layer** for real-time contextual voice and text AI agents.
- Built robust **AI integrations** with platforms like Salesforce, ServiceNow, and SAP, and 100+ others, unifying workflows.
- Enforced security using OAuth2.0 and JWT auth schemes, ensuring platform integrity via SAST security analysis.
- Developed **low-latency Voice AI** solutions in collaboration with **multiple Tata companies and MNCs** to streamline automated operations and enhance customer experience.

**Full Stack Developer Intern**, *Arxena Inc.* (arxena.com) June 2024 - July 2024

- Built a **GPT-powered WhatsApp recruitment chatbot** in **NestJS, GraphQL and WebSocket API** that screened candidates on JD-based questions, shared JD PDFs, stored resumes in **Amazon S3** and transcribed voice answers, with a WhatsApp chat UI in **React.js** inside **Twenty CRM** showing each candidate's CRM status.
- Reduced **AWS EC2** costs by up to **75%** by moving builds off the live server to on-demand instances from an **AMI** (details).

## OPEN SOURCE MENTORSHIP

**The Linux Foundation - LFX - Linux Kernel Mentorship Program mentee, Spring 2026** (link) March 2026 - August 2026

- Graduated with **11 patches** contributed to Linux Kernel mainline; published a blog sharing my experience.
- Converted text-based **Devicetree bindings** to JSON Schema; fixed documentation links, typos, and driver error handling.
- Worked with **kernel development workflow**, **patch submission**, and **review cycles** and used tools like **cscope**, **Coccinelle**, and **smatch** for static analysis. Selected for the mentorship program from **693 applicants**.

## PROJECTS

**Redspot**, *FastAPI, React.js, Clickhouse, Kafka, PostgreSQL, Docker* | (GitHub) June 2026

- Built a **real-time website click analytics platform** with a FastAPI ingestion server, Kafka message queue, and ClickHouse storage pipeline that captures user interactions via a lightweight JavaScript tracking script.
- Developed a React dashboard that renders target websites in an iframe and overlays a **color-coded heatmap visualization of click density** using canvas, powered by live data queried from ClickHouse

**SpeakSure**, *Python, FastAPI, Tensorflow, React.js, MongoDB* | (Live app, GitHub) October 2025

- Built an **AI behavioural interview platform** that scores **spoken confidence** (1 to 5) with Wav2Vec2 and an MLP, with transcription, speech rate, filler words and **Gemini feedback**, analysed by a **background job queue on MongoDB**.
- Moved inference from PyTorch and TensorFlow to **ONNX Runtime**, cutting about **3.5 GB** of ML dependencies from the server while running **19% faster** per chunk; models trained on a custom **751**-clip dataset (paper at ICCIS 2025).

**Shortomega**, *React.js, Next.js, Node.js, NestJS, TypeScript, Redis, Docker* | (Live app, GitHub) November 2024

- Developed a **Full Stack URL Shortening App** that generates short URLs for long URLs, with a focus on system design.
- Published a technical blog describing the methods of using **Redis as a Primary Database**.
- Implemented **Dashboard, Analytics data aggregation and Rate Limiting** features, and **Dockerized** the application.

## SKILLS

**Languages**: TypeScript, JavaScript, Python, Go, C, C++, Java
**Backend & APIs**: NestJS, FastAPI, Node.js, GraphQL, gRPC
**AI**: MCP, A2A, RAG, LangChain, LangGraph, Agentic AI
**Web**: React.js, Tailwind CSS, HTML, CSS, Next.js, HeroUI
**Data & Messaging**: PostgreSQL, MongoDB, Redis, Kafka
**Platforms & Tools**: Linux, Docker, Kubernetes, AWS

## RESEARCH AND PUBLICATION

**Evaluating Human Speech Confidence with Deep Learning Techniques**, **Ninad Naik, et al.** September 2025

- Validated **Multi Layer Perceptron**, **Fuzzy Hypersphere Neural Network** and **Logistic Regression** classifiers over **MFCC** and **Wav2Vec2** features on a custom **751-recording** dataset using **5-fold stratified cross-validation**, reaching **96% relaxed accuracy** across 5 confidence levels.
- Presented at **ICCIS 2025**, BITS Pilani, Goa; accepted for publication in SCOPUS-indexed **Springer** LNNS. (Certificate)

## ADDITIONAL INFORMATION

- Open Source Contribution to the Linux Kernel, n8n.io and CircuitVerse.
- First Runner Up (CSS Battle 2022 & Javagenix 2022), 2nd Runner Up (PyExpo 2023), Top 10 (ACE Hacks 2024)
- Maintaining a **tech blog** to share experiences, learnings and results on tech.ninadnaik.me